

Internal short circuit and thermal runaway evolution mechanism of fresh and retired lithium-ion batteries with LiFePO₄ cathode during overcharge

Cong-jie Wang^a, Yan-li Zhu^{a,*}, Fei Gao^b, Xin-ya Bu^a, Heng-shuai Chen^a, Ting Quan^a, Yi-bo Xu^a, Qing-jie Jiao^a

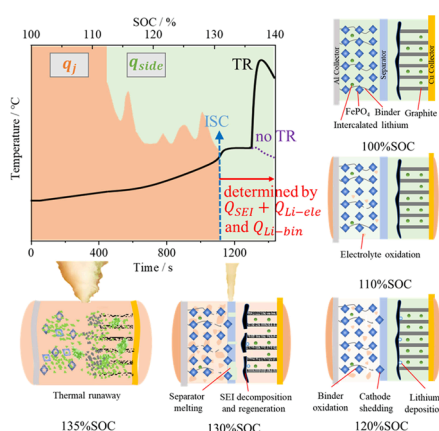
^a State Key Laboratory of Explosion Science and Technology, Beijing Institute of Technology, Beijing 100081, China

^b Battery Energy Storage Technology Laboratory, China Electric Power Research Institute, Beijing 100192, China

HIGHLIGHTS

- Overcharge-to-thermal-runaway evolution mechanism is analyzed.
- Joule heat plays a key role in the occurrence of internal short circuit.
- The effects of various side reactions on thermal runaway are decoupled.
- The 70% SOH (state of health) battery may be more prone to thermal runaway.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Retired lithium-ion battery
Thermal runaway mechanism
SOH
Overcharge
Safety

ABSTRACT

The safety evolution behavior of LiFePO₄/graphite batteries with different states of health (SOHs) under overcharge is studied based on material morphology, structure, thermal stability and heat analysis. The overcharge results of the 100 % SOH battery show that with increasing state of charge (SOC), the cathode material gradually falls off due to binder oxidation. The number of pores in the separator is significantly reduced when the SOC reaches 120 %, resulting in increased internal resistance. Before the internal short circuit (ISC), the degree of lithium intercalation in the anode increases, and the heat release of the reaction between lithiated graphite and the binder increases, whereas both decrease after ISC due to severe side reactions. The heat release from SEI decomposition increases after ISC as the SOH of the retired battery decreases, while the heat release from the reaction between lithiated graphite and binder decreases. The ISC starts from the positive collector side. Thermal analysis results show that Joule heat plays a key role in the occurrence of ISC. After ISC, Q_{SEI} (SEI decomposition heat) + Q_{Li-ele} (lithium and electrolyte reaction heat) and Q_{Li-bin} (lithiated graphite and binder reaction heat) together determine the difference in thermal runaway (TR) behavior of different SOH batteries.

* Corresponding author.

E-mail address: zhuyanli1999@bit.edu.cn (Y.-l. Zhu).

<https://doi.org/10.1016/j.apenergy.2022.120224>

Received 17 March 2022; Received in revised form 28 September 2022; Accepted 24 October 2022

Available online 3 November 2022

0306-2619/© 2022 Elsevier Ltd. All rights reserved.

1. Introduction

With the rapid popularization of new energy vehicles powered by lithium-ion batteries (LIBs) [1,2], it is expected that a large number of LIBs will be retired every year in the future. The reuse of retired LIBs is beneficial to maximize resource utilization [3]. However, the safety problem caused by thermal runaway (TR) of LIB is one of the main obstacles to their reuse. The internal and external characteristics of retired LIBs have changed compared with new batteries, including battery swelling, SEI thickening, and reduction of active lithium, etc. [4–6], which may bring new safety risks [7,8]. In addition, according to the LIB safety research report, electric vehicle safety accidents generally occur on vehicles that have been in operation for about a year [9]. Therefore, the TR-related safety problem of aged battery has attracted more and more attention. Overcharge is one of the main causes of battery TR [10], which may be caused by unbalanced charging of batteries in modules and failure of supporting devices [11,12]. Due to the additional input of electrical energy into the battery, overcharging can easily lead to fire and explosion of LIBs. Therefore, it is very necessary to study the evolution characteristics and failure mechanism of retired batteries under overcharge.

Researchers have carried out many studies on the TR behavior and mechanism of fresh LIBs under overcharge. Zhu et al. [13] studied the ignition mechanism of $\text{Li}[\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}]\text{O}_2/\text{graphite}$ (NCM111/C) batteries under overcharge through experiments in air/argon atmosphere, and revealed that the three elements of battery ignition are the combustibles injected by the battery, high-temperature electrode materials and oxygen from the air. Wang et al. [14] studied the TR differences of LIBs with different cathode materials (LiFePO_4 , NCM111, $\text{Ni}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ (NCM622) and $\text{Li}[\text{Ni}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}]\text{O}_2$ (NCM811)) based on the overcharge abuse test, and evaluated the safety of the batteries through thermal characteristics and other parameters. The results showed that the TR risk sequence was $\text{LFP} > \text{NCM811} > \text{NCM622} > \text{NCM111}$. In terms of TR mechanism, Ren et al. [11,15] studied the overcharge failure mechanism of NCM111- LiMn_2O_4 battery based on material characterization and thermo-electric coupling model. The results showed that side reactions such as electrolyte oxidation and Mn ion dissolution occurred at the cathode during overcharge, while lithium deposition occurred at the anode, reducing the thermal stability of the anode. Furthermore, separator melting was considered to be one of the key initiation points of TR. Qi et al. [16] studied the TR mechanism of NCM111/C batteries under overcharge based on Differential scanning calorimetry-Mass spectrometry (DSC-MS). The results showed that the thermal stability of cathode and anode decreased with the increase of SOC. In addition, the SEI decomposition was the onset of cell self-heating, while the formation of hydrogen fluoride was considered to be the onset of TR. Mao et al. [17] investigated the physical and chemical changes inside the NCM622 battery during overcharge. The results showed that the active material gradually decomposed at the cathode with the increase of SOC, and metallic Ni appeared. At the anode, the precipitated metallic Li reacted with H_2 to form LiH. Ultimately, LiH and metallic Ni accelerated side reactions and eventually led to battery TR. The current research helps to understand the TR process and TR triggering mechanism of new batteries. However, the TR evolution mechanism is different for different types of batteries. For example, the ISC is considered to be one of the key factors triggering battery TR during overcharge for the NCM battery. However, because the ISC and TR occur almost simultaneously during overcharge [13–19], it is difficult to study the ISC process independently. The LiFePO_4 battery to be studied in this paper is different from the NCM battery in that the ISC and TR do not occur at the same time, which is helpful to further understand the TR evolution mechanism of LIB.

There are only a few studies on the TR behavior of aged batteries. Our research group [20] studied the overcharge-to-TR features of fresh and retired LiFePO_4 batteries with 100 %–60 % SOH. The results showed that the battery TR became more serious with the increase of SOH. When

the battery was overcharged at 1C rate, the initial temperature of battery TR first decreased and then increased with the decrease of SOH, and it was the lowest at 70 % SOH. Feng et al. [21] studied the TR behavior of LIBs with 100 %–80 % SOH (NCM532/ LiCoO_2 blend cathodes). The aged batteries used were from the cycle aging of new batteries under 2C constant current and 4.5 V cut-off voltage. The results showed that the TR of batteries at different SOHs originated from the ISC caused by the melting of the separator. Based on the parameters such as temperature and voltage during overcharge, it was considered that the safety of aged battery was lower than that of new battery. Current research shows that TR also occurs in aged batteries, and the TR behavior is different from that of new batteries. However, the underlying mechanism of the difference in thermal behavior between aged batteries and fresh batteries during overcharge is still unclear, and there is still a lack of relevant research.

In this paper, the evolution behavior of material morphology, structure, and thermal stability of $\text{LiFePO}_4/\text{graphite}$ batteries at 100 % SOH during overcharge was firstly investigated. Further, the relationship between the material properties and the SOH (100 %, 75 %, 65 %, 55 %) was studied through the forced cooling experiment of liquid nitrogen after the battery ISC. Finally, through the combination of thermal calculation and measured thermal data, the heat generation behavior of the whole process before and after ISC was analyzed, and the thermal runaway evolution mechanism was revealed. The findings help to understand the safety of retired LIBs and provide guidance for the reuse of retired LIBs.

2. Experiment

2.1. Battery sample

The selected prismatic LIBs are from BYD, including fresh batteries (100 % SOH) and aged batteries (70–55 % SOH) retired from logistics vehicles. The SOH is an indicator of battery ageing, and the SOH is defined as follows [19].

$$\text{SOH} = \frac{\text{nominal capacity of aged cell}}{\text{nominal capacity of fresh cell (25 Ah)}} (100\%)$$

The nominal capacity of the fresh battery is 25Ah, the main components of the cathode and anode are lithium iron phosphate and graphite, and the cut-off voltage is 2.0–3.8 V.

2.2. Material acquisition of 100 % SOH battery at different SOC

The 100 % SOH battery was overcharged to 110 % SOC, 120 % SOC, ISC (ie voltage dip, ~130 % SOC), and TR (~135 % SOC) using 1C rate (25A), respectively. Here, a fully charged battery is defined as 100 % SOC [16], and the SOC increases by 10 % for every 6 min of overcharge using 1C rate. Fig. 1A and B show the temperature and voltage curves of the battery when overcharged to TR and ISC, respectively. The battery surface temperature was measured by a K-type thermocouple located at the center of the largest surface of the battery. It should be noted that when the battery was overcharged to about 130 % SOC, an ISC occurred in the battery. At this time, liquid nitrogen was immediately sprayed on the battery surface for cooling to slow down or terminate the reaction inside the battery, as shown in Fig. 1B. The purpose of cooling the battery was to quickly terminate the side reactions inside the battery, so as to analyze the ISC mechanism and battery material characteristics. When the battery was overcharged to about 135 % SOC, it was determined based on the temperature profile that TR had occurred, at which point the overcharge was terminated, as shown in Fig. 1A. Then, the cells with different SOC were disassembled in a glove box in an argon atmosphere, and the cathode and anode materials were scraped from the current collector and kept for testing. The schematic diagram of the experimental process is shown in Fig. 1C.

2.3. Material acquisition of different SOH batteries

The 75 %SOH, 65 %SOH, and 55 %SOH batteries were overcharged at 1C rate to an ISC, and then cooled by liquid nitrogen. The internal materials of LIBs with different SOHs were obtained by the same method in Section 2.2 and reserved for use, as shown in Fig. 1C.

2.4. Material characterization methods

Scanning electron microscopy (SEM) was used to analyze the microscopic morphology of the battery materials, X-ray diffraction (XRD) was used to analyze the crystal structure of the battery materials, and Thermogravimetry-Differential Scanning Calorimetry (TG-DSC) was used to analyze the thermal stability of the battery materials. The parameters of relevant instruments are shown in the previous articles of our research group [4,16].

3. Results and discussions

3.1. TR evolution behavior of 100 % SOH battery under overcharge

3.1.1. Morphology analysis

Fig. 2A-C shows the surface morphologies of cathode, separator, and anode with different SOC. Fig. 3 shows the corresponding SEM images. As shown in Fig. 2A, there was no obvious change in the cathode surface before 110 % SOC, while at 120 % SOC, obvious “bubbly” appeared on the cathode surface, which would lead to the loss of cathode active material. Part of the positive current collector (aluminum foil) was exposed after about 130 % SOC (after ISC), and the cathode active material was obviously peeled off, which was also confirmed by the microscopic morphology of the cathode. As shown in Fig. 3A, pits appeared on the cathode surface at 120 % SOC and 130 % SOC, and the active material fell off. The shedding of the cathode material may be

attributed to the oxidative decomposition of PVDF binder. As shown in Fig. 1B, the battery voltage had reached 5.9 V when the battery was overcharged to 120 % SOC. In addition, the electrode potential of lithium iron phosphate would also reach about 5 V during overcharge [22]. According to literature [23], when the potential reached 5 V, PVDF would gradually decompose, and the higher the potential, the faster the decomposition rate. Decomposition of the binder may generate gas that caused a “bulge” on the cathode surface. After 135 % SOC, the battery ran into TR. Under the action of high temperature, PVDF decomposed further and the cathode material fell off almost completely.

As shown in Fig. 2C, the yellow color of lithiated graphite was deeper with the increase of SOC from 100 % to 120 %, indicating that the degree of lithium intercalation in the anode increased gradually. However, the unevenness of lithium in the anode also gradually increased, and lithium deposition occurred locally [24]. When the battery was overcharged to 130 % SOC, a large number of black spots appeared on the anode surface, but the anode remained yellow as a whole. Therefore, it was considered that the battery had only a local ISC at about 130 % SOC. In addition, as shown in Fig. S1, the area on the positive collector side was damaged most seriously at 130 % SOC (after ISC), so the ISC may start from the positive collector side. This was because the impedance of the positive aluminum foil was higher than that of the negative copper foil, and the local high temperature caused the separator to melt. After ISC, the side reaction in the area where lithium dendrites are deposited on the anode is more serious, resulting in a large number of black spots. When overcharged to 135 % SOC, the battery ran into TR, and the yellow anode completely turned black, indicating that the lithium in the graphite was consumed. It should be noted that the anode material did not fall off significantly before 135 % SOC, which also proved that the high potential of the cathode played an important role in the oxidative decomposition of the binder PVDF. Although the lithium intercalation of the anode changed with the increase of SOC and the side reactions became more serious, the layered structure of graphite did not change

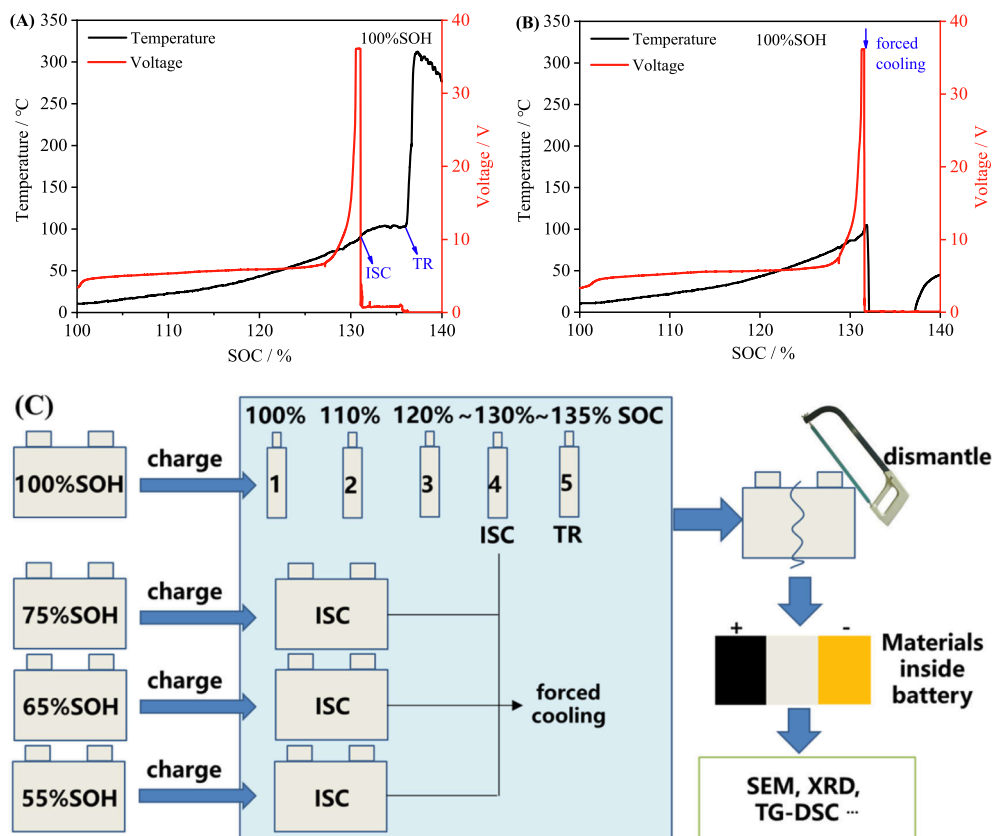


Fig. 1. Temperature – voltage vs SOC curves when overcharged to (A) TR and (B) ISC; (C) Schematic diagram of experimental operation.

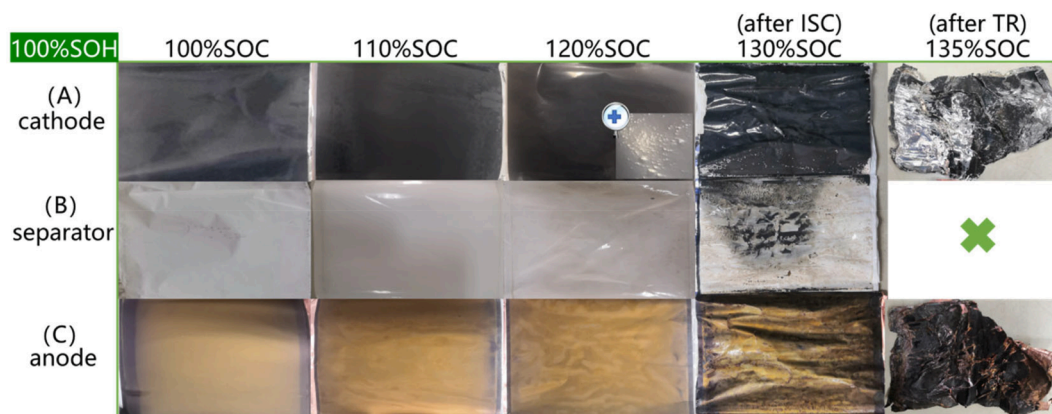


Fig. 2. Photos of cathode, separator and anode with various SOC levels for 100 %SOH batteries.

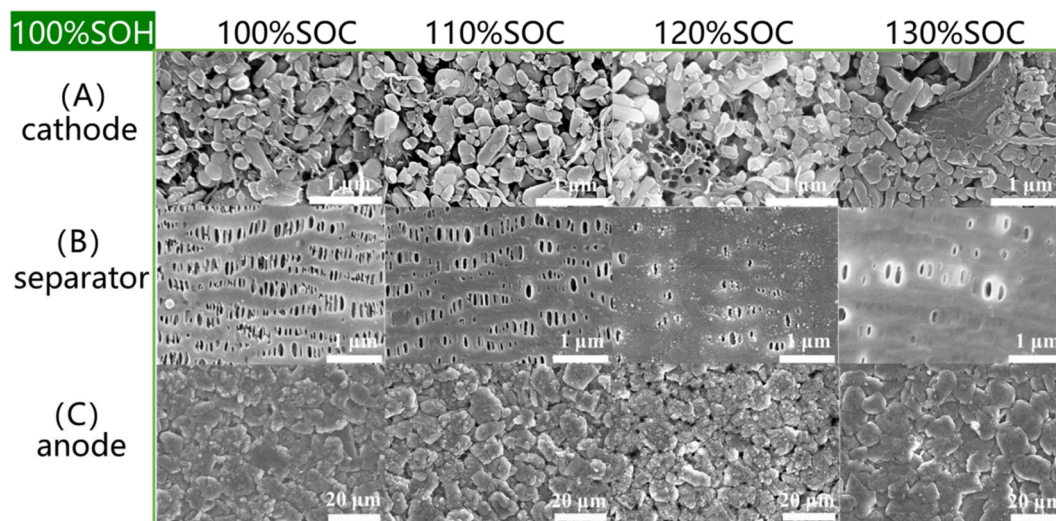


Fig. 3. SEM images of cathode, separator and anode with various SOC levels for 100 %SOH batteries.

significantly, as shown in Fig. 3C.

Fig. 2B shows the macroscopic morphology of the separator on the cathode side. More “impurities” were attached to the separator surface with the increase of SOC, which may be the products of the decomposition of the electrolyte and the binder. These substances on the separator can block the pores in the separator. As shown in Fig. 3B, the pores in the separator were significantly reduced at 120 % SOC, which increased the resistance of lithium ions passing through the separator.

3.1.2. Crystal structure analysis

Fig. 4 shows the XRD curves of anode and cathode with different SOC levels. As shown in Fig. 4A, the characteristic peaks of LiC_6 and LiC_{12} both increased with the increase of SOC from 100 % to 110 %, indicating that the degree of lithium intercalation in the anode increased. The (001) and (002) characteristic peaks of LiC_6 increased significantly with SOC increasing from 110 % to 120 %, while the characteristic peaks of LiC_{12} decreased, indicating that the main reaction at this stage was the

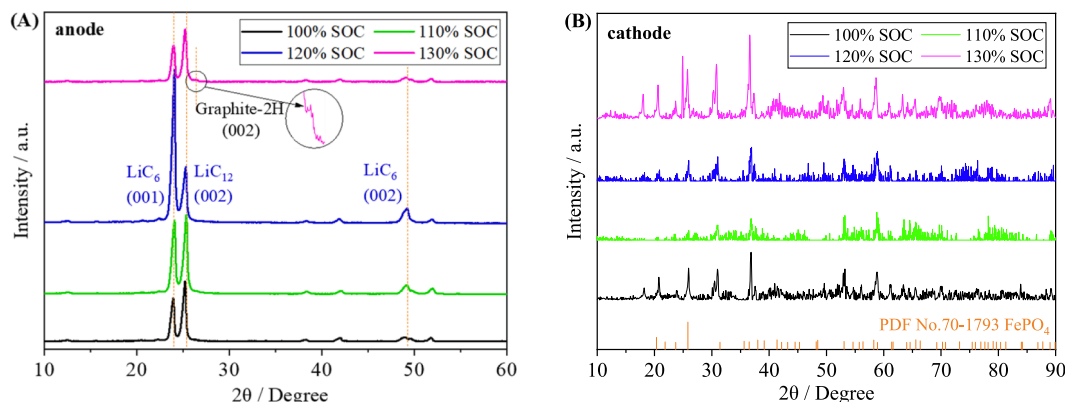


Fig. 4. XRD patterns of anode and cathode with various SOC levels for 100 %SOH batteries.

transformation from LiC_{12} to LiC_6 , and the lithium degree of graphite was further increased. When SOC reached 130 %, due to the rapid lithium removal of anode during ISC, the characteristic peak of LiC_6 and LiC_{12} decreased, and a small Graphite-2H characteristic peak appeared. However, it should be noted that there was still a large amount of lithium in the graphite after ISC, indicating that the ISC only caused the release of a small amount of electrical energy.

As can be seen from Fig. 4B, the cathode material exhibited complex diffraction peaks, indicating that it contained various impurities. The position of the characteristic peak hardly changed with the increase of SOC. Since the main active material of the cathode is LiFePO_4 , we believe that the active material after overcharge is mainly FePO_4 . However, as shown in Fig. 4B, there were still some diffraction peaks that did not match the standard PDF card of FePO_4 . Through phase search, we found that these extra diffraction peaks may be related to (Mn, Fe) PO_4 , LiFePO_4 , $\text{FePO}_4 \cdot 2\text{H}_2\text{O}$ and other substances. Through the elemental analysis (ICP and EDS) of cathode material with 100 % and 130 % SOC, the cathode material still contained a small amount of Mn and Li elements, as well as a significant excess of O element. As shown in Table 1, the molar ratio of P, Fe, Mn, Li and O at 100 % SOC is 1: 0.9030: 0.0006: 0.1186: 5.2090, while the molar ratio is 1: 0.9372: 0.0008: 0.0342: 5.2569 at 130 % SOC. Assuming that x in Li_xFePO_4 is 1 at 0 % SOC and 0.1186 at 100 % SOC, x is 0.0342 at 130 % SOC. Therefore, as the SOC increases from 100 % to 130 %, the SOC increases by 30 %, while the relative content of lithium in the cathode decreases by only 9.6 %. Therefore, serious side reactions occurred on the cathode side during overcharge, including the oxidative decomposition of electrolyte and binder.

3.1.3. Thermal stability analysis

Fig. 5A and B show TG-DSC curves of anode and cathode with different SOCs respectively. Table 1 shows the peak temperature and heat release / absorption corresponding to the anode DSC curve in Fig. 5A. It should be noted that the baseline is not deducted from the DSC curve, because this paper mainly focuses on the correlation analysis of the peak position and peak area in the DSC curve, rather than the absolute value in the curve. As shown in the DSC curves in Fig. 5A, the anode material exhibited one endothermic peak and three exothermic peaks. The peak 1# was an endothermic peak with a peak temperature of 69 °C at 100 % SOC, which was caused by the decomposition of LiPF_6 [4]. The peak temperature of peak 1# did not change significantly with the increase of SOC. However, the endothermic value corresponding to the peak 1# changed with increasing SOC due to the influence of the amount of LiPF_6 in the sample material. The peak 2# was an exothermic peak with a peak temperature of 112 °C at 100 % SOC, which was caused by SEI decomposition on the anode surface [25]. Due to the rupture and regeneration of SEI film on the anode surface [16], the heat release corresponding to peak 2# first decreased and then increased with the increase of SOC. The peak 3# was an exothermic peak with a peak temperature of 174 °C at 100 % SOC, which may be caused by the reaction between lithium plating and organic solvent [26]. The peak temperature of peak 3# did not change significantly with the increase of SOC. However, the heat release corresponding to the peak 3# increased from 24.5 $\text{J} \cdot \text{g}^{-1}$ to 59.1 $\text{J} \cdot \text{g}^{-1}$ with the SOC increasing from 100 % to 110 % due to the increase of lithium plating. Between 120 % and 130 % SOC, although the lithium plating continued to increase, the electrolyte and lithium were also continuously consumed, so the heat release did not change obviously. The peak 4# was an exothermic peak with a peak

temperature is 276 °C at 100 % SOC, which was caused by the reaction of lithiated graphite and the binder [27,28], so the reaction was closely related to the degree of lithium intercalation in graphite. With the increase of SOC from 100 % to 120 %, the heat release of peak 4# increased significantly from 102.6 $\text{J} \cdot \text{g}^{-1}$ to 228.5 $\text{J} \cdot \text{g}^{-1}$. However, when the SOC increased from 120 % to 130 %, the exothermic heat for the peak 4# decreased to 182.0 $\text{J} \cdot \text{g}^{-1}$ due to the decrease of the lithium content in the graphite caused by the ISC. The DSC results and the XRD results in Section 3.1.2 corroborate each other.

As shown in the DSC curve in Fig. 5B, the cathode material with various SOCs exhibited an obvious endothermic peak at 173–194 °C, which may be the decomposition of the electrolyte attached to the cathode under the action of the active material, accompanied by the generation of gases such as CO and CH_4 [4]. The peak temperature of the endothermic peak increased with the increase of SOC from 100 % to 120 %, while the peak temperature decreased when the SOC reaches 130 %. The oxidation of some electrolyte changed the composition ratio of the electrolyte during overcharge, resulting in the change of peak temperature.

The TG curve in Fig. 5A can reflect the gas production of the side reaction. As shown in Fig. 5A, the SEI decomposition and the reaction between lithium plating and electrolyte are the main reasons for anode weight loss, while the reaction between the lithium intercalated graphite and the binder has almost no weight loss. Under the action of high temperature, the metastable components ($(\text{CH}_2\text{OCO}_2\text{Li})_2$, etc.) in SEI film decomposed and produced gases such as CO_2 and O_2 , and the reaction between lithium and electrolyte mainly produced alkane gases such as CH_4 , C_2H_4 , etc. [29]. The relative molecular weight of these gases was large, resulting in significant weight loss of anode materials. However, the reaction between lithium and binder mainly produced H_2 with low relative molecular weight [29], so there was almost no weight loss on the TG curve. It can be seen from Fig. 5A that the anode weight loss hardly changed as the SOC increases from 100 % to 120 %. However, the weight loss of the anode increased significantly by 14.0 % when the SOC increased to 130 %, which may be due to the production of more gas with large molecular weight by side reactions. According to the TG curve in Fig. 5B, the weight loss of the cathode increased with increasing SOC, indicating that more mass of gas was produced. The weight loss of 130 % SOC cathode reached 13.9 % at 300 °C. To sum up, both the cathode and the anode generated a large amount of gas during overcharge, which contributed to the explosion of the battery.

3.2. Effects of SOH on TR evolution behavior under overcharge

Fig. 6 shows the photos of cathode, separator, and anode with different SOHs after ISC. Fig. S2 shows the corresponding SEM images. The cyclic de-intercalation of lithium made the cathode material expand and deform during battery aging, and the performance of binder decreased. Therefore, as shown Fig. S2, the cracks in the cathode material increased as the SOH decreased, which would cause the loss of cathode active material. The separator was seriously blocked with the decrease of SOH, and the pores in the separator were no longer observed after ISC for the aged battery (75 %–55 %SOH, Fig. S2), which increased the internal resistance of the aged battery. The black area on the anode surface gradually increased with the decrease of SOH (Fig. 6), indicating that the lithium content in graphite decreased. This was further supported by XRD results. As shown in Fig. 4 and Fig. S3, the anode with 100 % SOH was mainly LiC_6 and LiC_{12} , while the anode with 75 % SOH was mainly LiC_{12} and contained a small amount of LiC_6 . The anode with 65 % SOH was mainly LiC_{12} and a small amount of Graphite-2H, while the anode with 55 % SOH was mainly Graphite-2H. On the one hand, the loss of active lithium was one of the main causes of battery aging [9,30]. Therefore, with the decrease of SOH from 100 % to 55 %, a large amount of active lithium was lost in the battery due to the side reactions such as the decomposition and regeneration of SEI. On the other hand, the anode material became more “loose and porous” after battery aging, and

Table 1

Molar ratios of cathode elements measured by ICP and EDS for 100 %SOH batteries.

SOC	P- (ICP and EDS)	Fe-(ICP)	Mn-(ICP)	Li-(ICP)	O-(EDS)
100 %	1	0.9030	0.0006	0.1186	5.2090
130 %	1	0.9372	0.0008	0.0342	5.2569

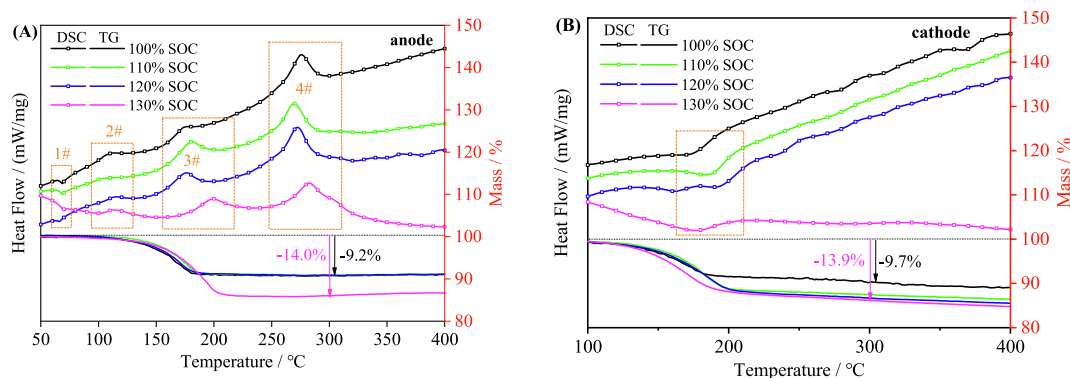


Fig. 5. TG-DSC curves of anode and cathode with various SOC for 100 %SOH batteries.

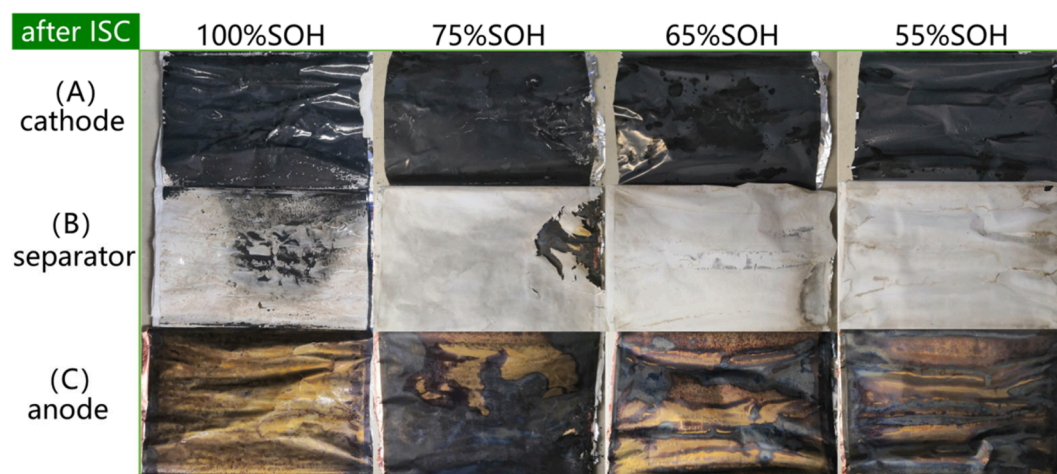


Fig. 6. Photos of cathode, separator and anode with various SOHs after ISC.

the gap of the layered structure increased (Fig. S2). When the battery was overcharged to ISC, the lithium embedded in graphite and the lithium deposited on the anode surface were more likely to react with the electrolyte, further reducing the active lithium.

Fig. 7A shows the TG-DSC curves of the anode with different SOHs. Table 3 shows the peak temperature and heat release/absorption in the DSC curve in Fig. 7A. As shown in Fig. 7A and Table 3, the decomposition (peak 1#) of LiPF_6 had no obvious relationship with SOH. However, the heat generation of SEI decomposition (peak 2#) increased gradually from $11.5 \text{ J} \cdot \text{g}^{-1}$ of 100 % SOH to $97.6 \text{ J} \cdot \text{g}^{-1}$ of 55 % SOH with the decrease of SOH. This was because the SEI film thickened during battery aging due to continuous rupture and repair of SEI [8,31]. SEI

decomposition played an important role in the early stage of TR development. The results in Table 3 show that the battery was more likely to accumulate heat in the early stage of TR with the decrease of SOH. As shown in Table 3, compared with the new battery (100 %SOH), the peak temperature and heat release of the reaction between deposited lithium and electrolyte (peak 3#) of aged batteries decreased. For aged batteries, there may be less deposited lithium at the anode during overcharge than new batteries due to the reduction of active lithium, so the heat release of the reaction decreased. The peak temperature of the reaction between lithium graphite and binder (peak 4#) increased with the decrease of SOH of aged battery, indicating that the reaction was more difficult to occur. However, the corresponding heat release decreased

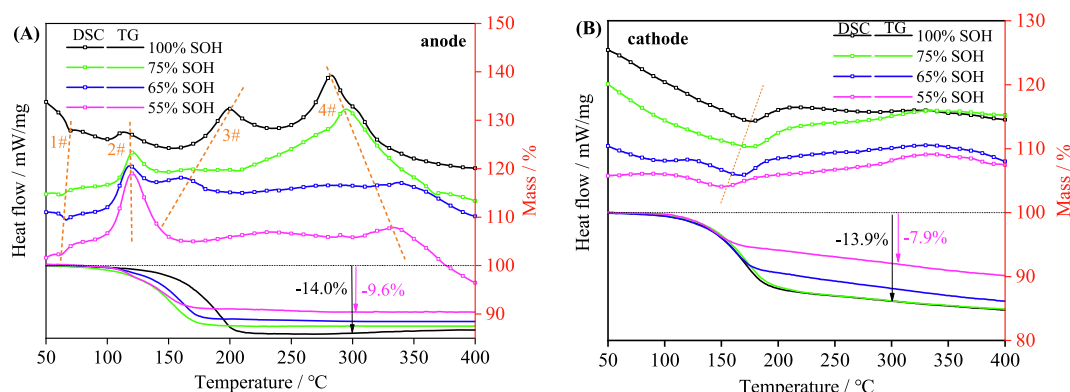


Fig. 7. TG-DSC curves of anode and cathode with different SOHs.

from 259.0 J·g⁻¹ of 75 % SOH to 51.2 J·g⁻¹ of 55 % SOH due to the decrease of lithium content in graphite. It should be noted that the heat release for 100 % SOH was lower than 75 % SOH, because the area without ISC was selected during the test for 75 % SOH. As shown in Fig. 6, the ISC of 100 % SOH battery occurred on the whole anode sheet in a point distribution, while the ISC of 75 % SOH was a concentrated regional distribution. The lithium content was higher where there was no ISC for 75 % SOH, so the measured heat release was large. In general, it can be concluded that with the decrease of SOH, the reaction between lithium intercalated graphite and binder was more difficult to occur, and the heat release decreased, which may lead to the decrease of the maximum temperature after the battery TR. Fig. 7B shows the TG-DSC curves of cathode with different SOHs. It can be seen from the figure that with the decrease of SOH, the peak temperature of endothermic peak showed a downward trend due to the change of the proportion of electrolyte composition during battery aging, indicating that the reaction may be more prone to occur.

According to the TG curve in Fig. 7, the weight loss of anode and cathode increased with the increase of SOH. Combined with our previous test [19], the higher the SOH, the more likely the battery shell was to explode when the battery ran into TR. Therefore, it can be inferred that the higher the SOH, the greater the molar amount of gas produced by the battery TR.

3.3. TR evolution mechanism based on heat generation analysis

The heat accumulation during overcharge is the root cause of battery TR. It can be seen from Fig. 1A that the starting time of ISC of the LiFePO₄ battery was significantly earlier than that of TR, while the ISC and TR of the NCM battery occurred almost simultaneously [15,17,19]. Therefore, the heat generation analysis in this paper was divided into two parts: before and after ISC.

The total heat accumulation (Q) of the battery before ISC during overcharge includes Joule heat (Q_j), reversible entropy heat, side reaction heat (Q_{side}), and heat dissipation (Q_p). Since the Joule heat during overcharge is much greater than the reversible entropy heat, the reversible entropy heat can be ignored [11]. Therefore, Q can be approximately expressed by Eq. (1).

$$Q = Q_j + Q_{side} + Q_p \quad (1)$$

Based on the temperature change of the battery during overcharge, Q can be calculated from Eq. (2), where $M = 0.782$ Kg is the total mass of the battery. C_p is the specific heat capacity of the battery, taking 1100 J kg⁻¹ K⁻¹.

$$Q = MC_p \frac{dT}{dt} \quad (2)$$

Q_j can be calculated from the internal resistance (r_{bat}) of the battery during overcharge, as shown in Eq. (3), where I is the overcharge current.

$$Q_j = I^2 \int r_{bat} dt \quad (3)$$

The r_{bat} was measured experimentally. The test method was similar to the literature [11]. The overcharge current I used during the test was 12.5A (0.5C), with a 10 s rest for every 6 min of charging, as shown in Fig. 8. After obtaining the relationship between r_{bat} and SOC, the variation curve of r_{bat} with SOC (100 % < SOC < 131.1 %) was further obtained by polynomial fitting of the scatter plot, as shown in Eq. (4).

$$r_{bat} = 155.1876 - 551.54085 \cdot SOC + 737.184 \cdot SOC^2 - 439.0101 \cdot SOC^3 + 98.28274 \cdot SOC^4 \quad (4)$$

The total heat accumulation (Q) and Joule heat (Q_j) during overcharge were calculated based on Eq. (2) and (3), and the results are

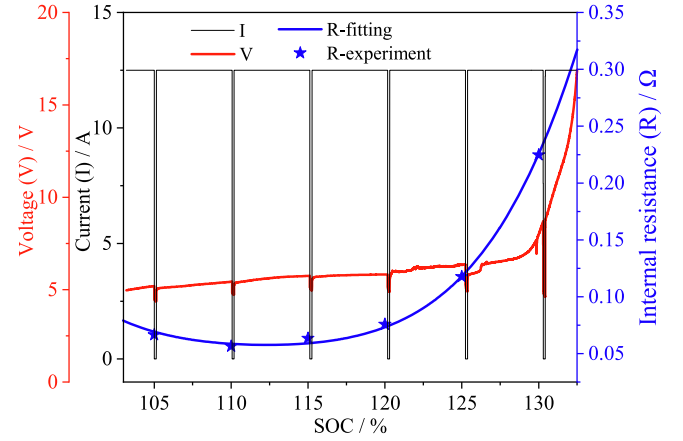


Fig. 8. Measurement of the internal resistance.

shown in Fig. 9. It can be seen from the Fig. 9 that before 814 s, the Joule heat (Q_j) is greater than the total heat accumulation (Q) due to heat dissipation. Q_j is only slightly smaller than Q after 814 s, indicating that Joule heat is an important heat source before ISC.

As shown in Fig. 9, the total heat accumulation (Q) increased linearly with the charging time in the early overcharge period (before 450 s), which was consistent with the results in the literature [14]. In fact, the side reaction hardly occurred in the early stage of overcharge, so it can be considered that $Q_j + Q_p$ was approximately proportional to the charging time. To evaluate the heat of side reaction (Q_{side}), it was assumed that $Q_j + Q_p$ was still proportional to the charging time after 450 s, as shown by the dotted line in Fig. 9A. Therefore, Q_{side} can be calculated by Eq. (5), and the result was shown in Fig. 9A.

$$Q_{side} = Q - (Q_j + Q_p) \quad (5)$$

Further, in order to compare the contribution of side reaction heat and Joule heat during overcharge, the heat generation powers q , q_j , and q_{side} corresponding to Q , Q_j , and Q_{side} were calculated based on Eq. (6), and the results were shown in Fig. 9B. As shown in Fig. 9B, Joule heat was the main heat source for most of the time before the ISC and played an important role in the temperature rise of the battery. The heat generation power of the side reaction gradually increased after 450 s, which cannot be ignored. When the battery was overcharged to about 1092 s, the heat generation power of the side reaction was greater than that of the internal resistance, and then the ISC occurred at 1119 s.

$$Q_i = \int q_i dt \quad (6)$$

The evolution process of the ISC can be speculated combined with the analysis in Section 3.1. The voltage and Joule heat increased rapidly before ISC, resulting in a further increase in the battery temperature. Especially on the positive collector side, Joule heat was greater. In addition, a large amount of metal lithium was deposited on the anode surface before ISC. Under the action of high temperature, the heat generation power of side reactions increased and exceeded the Joule heat power when approaching the ISC. Under the action of Joule heat and side reaction heat, the ISC was finally caused on the positive collector side and gradually extended to other areas of the battery. Although it was uncertain which heat source ultimately triggered the battery's ISC, the contribution of Joule heat from the positive collector side was undoubtedly significant.

After the battery's ISC, the Joule heat (Q_j) decreased sharply, and the heat generation from side (Q_{side}) reactions became the main heat source. According to the DSC test in Section 3.1 and 3.2, the heat generated by SEI decomposition (Q_{SEI} represents Q_2 in Tables 2 and 3) and the heat generated by the reaction between lithium and electrolyte (Q_{Li-ele} represents Q_3 in Tables 2 and 3) were the key to trigger battery TR. The heat

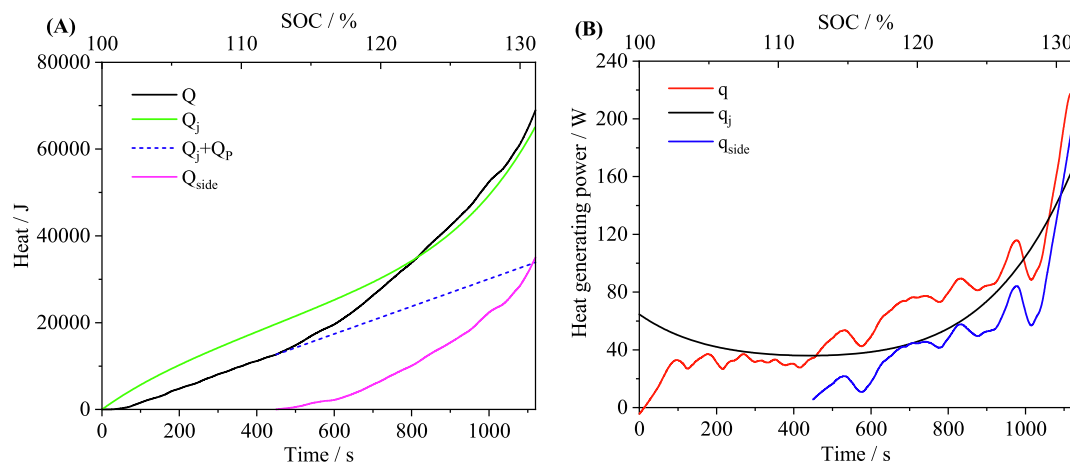


Fig. 9. (A) Heat (B) heat generation power – time/SOC curves during overcharge.

generated by the reaction between lithiated graphite and the binder (Q_{Li-bin} represents Q_4 in Tables 2 and 3) was an important reason for the sharp rise in battery temperature. However, the occurrence of battery TR, the TR trigger temperature and the time from ISC to TR were all related to Q_{SEI} , Q_{Li-ele} and Q_{Li-bin} . The larger the value of $Q_{SEI} + Q_{Li-ele}$ and Q_{Li-bin} , the easier it was to trigger the battery TR. Based on the relationship between Q_{SEI} , Q_{Li-ele} , Q_{Li-bin} and SOH in Section 3.2, $Q_{SEI} + Q_{Li-ele}$ decreased with the decrease of SOH from 100 % to 75 %, while Q_{Li-bin} increased. As the SOH decreased from 75 % to 55 %, $Q_{SEI} + Q_{Li-ele}$ increased, while Q_{Li-bin} decreased. Therefore, it can be speculated that within 100 %-55 % SOH, there was a certain SOH whose TR is most likely to be triggered. According to our previously published article [19], the TR onset temperature of 70 % SOH was the lowest when the batteries at 100 %-60 %SOH were overcharged using a 1C-rate, which also confirms the above view.

4. Conclusions

In this paper, the TR evolution process of 100 % SOH battery during overcharge was firstly analyzed by characterizing the materials (SEM, XRD and TG-DSC, etc.) inside the battery with different SOC. Furthermore, the effect of SOH on TR evolution behavior was analyzed. Finally, the TR mechanism of LIBs with different SOHs was discussed based on the heat generation analysis.

The results show that side reactions such as binder oxidation on the cathode side became more severe with increasing SOC, causing the active material to fall off gradually. In addition, the cathode weight loss caused by side reactions increased with SOC. The number of pores in the separator was significantly reduced after 120 % SOC, resulting in an increase in internal resistance. With the increase of SOC from 100 % to 120 %, the lithium intercalation in the anode increased, resulting in the exothermic heat of the reaction between lithiated graphite and binder increasing from 102.6 J·g⁻¹ to 228.5 J·g⁻¹. In addition, the heterogeneity of lithium embedded in anode increased and lithium deposition gradually occurred. However, an ISC occurred in the battery at 130 %

Table 2

Peak temperature and heat of anode with different SOC for 100 %SOH batteries.

SOC	Peak 1# / °C	Q_1 / J·g ⁻¹	Peak 2# / °C	Q_2 / J·g ⁻¹	Peak 3# / °C	Q_3 / J·g ⁻¹	Peak 4# / °C	Q_4 / J·g ⁻¹
100	69	-2.3	112	11.0	174	24.5	276	102.6
110	69	-2.0	105	5.7	181	59.1	269	142.2
120	66	-0.9	117	6.3	176	51.2	273	228.5
130	70	-2.3	113	11.5	199	56.3	283	182.0

Table 3

Peak temperature and heat of anode with different SOHs.

SOH	Peak 1# / °C	Q_1 / J·g ⁻¹	Peak 2# / °C	Q_2 / J·g ⁻¹	Peak 3# / °C	Q_3 / J·g ⁻¹	Peak 4# / °C	Q_4 / J·g ⁻¹
100	70	-2.3	113	11.5	199	56.3	283	182.0
75	62	-1.8	121	25.0	167	4.7	294	259.0
65	67	-3.1	118	32.6	160	12.1	337	46.3
55	62	-1.3	120	97.6	–	–	334	51.2

SOC, and serious side reaction resulted in the reduction of lithium embedding in the anode. TG results showed that both the cathode and anode with 130 %SOC lost weight up to 14 %, indicating that a large amount of gas was produced.

The heat released by SEI decomposition increased with decreasing SOH due to the thickening of SEI, while the heat released by the reaction between lithium and binder decreased due to the decrease in the lithium content of the anode. Thermal analysis results showed that Joule heat was the main heat source for most of the time before ISC, which contributed a key role to the occurrence of ISC. When approaching ISC, the heat generation power of side reactions exceeded that of internal resistance, and the side reaction heat determined the whole TR process of LIBs with various SOHs after ISC. The larger the value of $Q_{SEI} + Q_{Li-ele}$ and Q_{Li-bin} , the more prone to TR and the higher the risk of the battery.

These results provide a direction for the development of safer lithium-ion batteries from internal materials, and also provides guidance for the targeted development of battery safety management technologies in the full life cycle.

CRediT authorship contribution statement

Cong-jie Wang: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft. **Yan-li Zhu:** Conceptualization, Methodology, Supervision, Resources. **Fei Gao:** Methodology, Investigation. **Xin-ya Bu:** Writing – review & editing. **Heng-shuai Chen:** Investigation, Writing – review & editing. **Ting Quan:** Supervision, Writing – review & editing. **Yi-bo Xu:** Investigation. **Qing-jie Jiao:** Supervision, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was financially supported by National Natural Science Foundation of China (Grant Nos. 52022013 and 51974031)

References

- [1] Chen J, Ren D, Hsu H, Wang Li, He X, Zhang C, Feng X, Ouyang M. Investigating the thermal runaway features of lithium-ion batteries using a thermal resistance network model. *App Energy* 2021;295:117038.
- [2] Cao J, Ling Z, Lin S, He Y, Fang X, Zhang Z. Thermochemical heat storage system for preventing battery thermal runaway propagation using sodium acetate trihydrate/expanded graphite. *Chem Eng J* 2022;433:133536.
- [3] Xu X, Mi J, Fan M, Yang K, Wang H, Liu J, et al. Study on the performance evaluation and echelon utilization of retired LiFePO₄ power battery for smart grid. *J Clean Prod* 2019;213:1080–6.
- [4] Wang C, Zhu Y, Gao F, Wang K, Zhao P, Meng Q, et al. Morphology, Structure, and Thermal Stability Analysis of Aged Lithium-Ion Battery Materials. *J Electrochem Soc* 2020;167:140550.
- [5] Han X, Lu L, Zheng Y, Feng X, Li Z, Li J, et al. A review on the key issues of the lithium ion battery degradation among the whole life cycle. *eTransportation* 2019; 1:100005.
- [6] Scipioni R, Jørgensen PS, Stroe DI, Younesi R, Simonsen SB, Norby P, et al. Complementary analyses of aging in a commercial LiFePO₄/graphite 26650 cell. *Electrochim Acta* 2018;284:454–68.
- [7] Xie S, Ren L, Yang X, Wang H, Sun Q, Chen X, et al. Influence of cycling aging and ambient pressure on the thermal safety features of lithium-ion battery. *J Power Sources* 2020;448:227425.
- [8] Abada S, Petit M, Lecocq A, Marlair G, Sauvart-Moynot V, Huet F. Combined experimental and modeling approaches of the thermal runaway of fresh and aged lithium-ion batteries. *J Power Sources* 2018;399:264–73.
- [9] Ren D, Hsu H, Li R, Feng X, Guo D, Han X, et al. A comparative investigation of aging effects on thermal runaway behavior of lithium-ion batteries. *eTransportation* 2019;2:100034.
- [10] Long M, Duan P, Gao Y, Wang X, Wu G, Wang Y. Boosting safety and performance of lithium-ion battery enabled by cooperation of thermotolerant fire-retardant composite membrane and nonflammable electrolyte. *Chem Eng J* 2022;432: 134394.
- [11] Ren D, Feng X, Lu L, Ouyang M, Zheng S, Li J, et al. An electrochemical-thermal coupled overcharge-to-thermal-runaway model for lithium ion battery. *J Power Sources* 2017;364:328–40.
- [12] Chen Y, Kang Y, Zhao Y, Wang L, Liu J, Li Y, et al. A review of lithium-ion battery safety concerns: The issues, strategies, and testing standards. *J Energy Chem* 2021; 59:83–99.
- [13] Zhu Y, Wang C, Gao F, Shan M, Zhao P, Meng Q, et al. Rupture and combustion characteristics of lithium-ion battery under overcharge. *J Energy Storage* 2021;38: 102571.
- [14] Wang Z, Yuan J, Zhu X, Wang H, Huang L, Wang Y, et al. Overcharge-to-thermal-runaway behavior and safety assessment of commercial lithium-ion cells with different cathode materials: A comparison study. *J Energy Chem* 2021;55:484–98.
- [15] Ren D, Feng X, Lu L, He X, Ouyang M. Overcharge behaviors and failure mechanism of lithium-ion batteries under different test conditions. *App Energy* 2019;250:323–32.
- [16] Qi C, Zhu Y-I, Gao F, Yang K, Jiao Q-J. Morphology, Structure, and Thermal Stability Analysis of Cathode and Anode Material under Overcharge. *J Electrochem Soc* 2018;165(16):A3985–92.
- [17] Mao N, Zhang T, Wang Z, Cai Q. A systematic investigation of internal physical and chemical changes of lithium-ion batteries during overcharge. *J Power Sources* 2022;518:230767.
- [18] Zhu X, Wang Z, Wang Y, Wang H, Wang C, Tong L, et al. Overcharge investigation of large format lithium-ion pouch cells with Li(Ni_{0.6}Co_{0.2}Mn_{0.2})O₂ cathode for electric vehicles: Thermal runaway features and safety management method. *Energy* 2019;169:868–80.
- [19] Yuan Q, Zhao F, Wang W, Zhao Y, Liang Z, Yan D. Overcharge failure investigation of lithium-ion batteries. *Electrochim Acta* 2015;178:682–8.
- [20] Wang C-J, Zhu Y-I, Gao F, Qi C, Zhao P-L, Meng Q-F, et al. Thermal runaway behavior and features of LiFePO₄/graphite aged batteries under overcharge. *Int J Energy Res* 2020;44(7):5477–87.
- [21] Feng L, Jiang L, Liu J, Wang Z, Wei Z, Wang Q. Dynamic overcharge investigations of lithium ion batteries with different state of health. *J Power Sources* 2021;507: 230262.
- [22] Fedorova AA, Anishchenko DV, Beletskii EV, Kalnin AY, Levin OV. Modeling of the overcharge behavior of lithium-ion battery cells protected by a voltage-switchable resistive polymer layer. *J Power Sources* 2021;510:230392.
- [23] Xiao Q, Wang X, Li W, Li Z, Zhang T, Zhang H. Macroporous polymer electrolytes based on PVDF/PEO-b-PMMA block copolymer blends for rechargeable lithium ion battery. *J Membrane Sci* 2009;334(1-2):117–22.
- [24] Lu W, López CM, Liu N, Vaughey JT, Jansen A, Dennis W. D. Overcharge Effect on Morphology and Structure of Carbon Electrodes for Lithium-Ion Batteries. *J Electrochem Soc* 2012;159(5):A566–70.
- [25] Pasquier AD, Disma F, Bowmer T, Gozdz AS, Amatucci G, Tarascon J-M. Differential Scanning Calorimetry Study of the Reactivity of Carbon Anodes in Plastic Li-Ion Batteries. *J Electrochem Soc* 1998;145(2):472–7.
- [26] Spotnitz R, Franklin J. Abuse behavior of high-power, lithium-ion cells. *J Power Sources* 2003;113(1):81–100.
- [27] Oka H, Makimura Y, Uyama T, Nonaka T, Kondo Y, Okuda C. Changes in the stage structure of Li-intercalated graphite electrode at elevated temperatures. *J Power Sources* 2021;482:228926.
- [28] Zheng Y, Shi Z, Ren D, Chen J, Liu X, Feng X, et al. In-depth investigation of the exothermic reactions between lithiated graphite and electrolyte in lithium-ion battery. *J Energy Chem* 2022;69:593–600.
- [29] Wang Q, Ping P, Zhao X, Chu G, Sun J, Chen C. Thermal runaway caused fire and explosion of lithium ion battery. *J Power Sources* 2012;208:210–24.
- [30] Zhu X, Wang Z, Wang C, Huang L. Overcharge Investigation of Large Format Lithium-Ion Pouch Cells with Li(Ni_{0.6}Co_{0.2}Mn_{0.2})O₂ Cathode for Electric Vehicles: Degradation and Failure Mechanisms. *J Electrochem Soc* 2018;165(16): A3613–29.
- [31] Liu P, Wang J, Hicks-Garner J, Sherman E, Soukiazian S, Verbrugge M, et al. Aging mechanisms of LiFePO₄ batteries deduced by electrochemical and structural analyses. *J Electrochem Soc* 2010;157:A499–507.